

This effectively doubled the frequency of the memory; DDR 266 memory actually works at 133 MHz. It is important to note that the term is "DDR 266" and not "DDR 266 MHz." Another mode of notation is by referring to the peak data transfer speeds. So a DDR 266 module is also referred to as PC 2100, since it can transfer 2100 Megabytes per second. DDR modules have 184 contacts, and are not backward-compatible.

1.2.4 DDR2 SDRAM

An evolution of DDR SDRAM, DDR2 allows four data transfers per clock cycle, by clocking the internal bus at twice the speed of

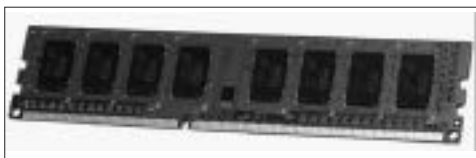


Screenshot 13: DDR module.jpeg

the memory clock. Therefore the effective frequency of the memory becomes 4 times its actual frequency. A DDR2 800 module operates at 200 MHz. As in the case of DDR SDRAM, the alternate notation relying on the maximum data transfer speeds is also used. PC2 3200 refers to DDR2 400. A DDR2 module has 240 contacts, and is not backward-compatible.

1.2.5 DDR3 SDRAM

This is the latest iteration of SDRAM, and increases the internal bus speed to 8 times the memory clock, effectively operating at 8 times



Screenshot 15: DDR3 module.jpeg

the frequency. DDR3 800 operates at 100 MHz and is also referred to as PC3 6400. DDR3 is still a cutting-edge technology, and is supported by very few motherboards and CPUs. DDR3 modules also have 240 pins, but they are keyed differently, so they cannot be inserted into a DDR2 slot.